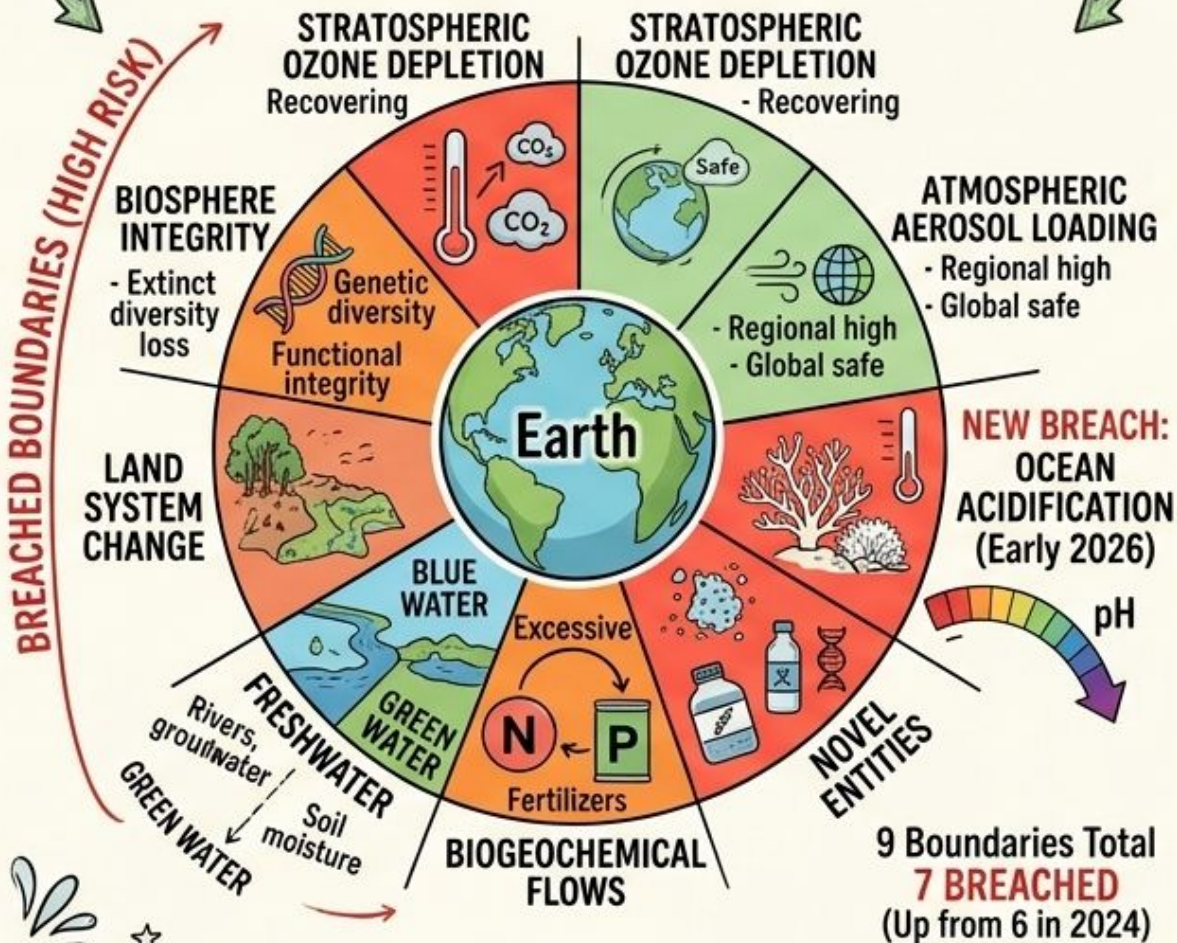


UPSC CSE PRELIMS 2026: ENVIRONMENT & ECOLOGY CURRENT TOPICS

STATE OF INDIA'S ENVIRONMENT (SOE) 2026: KEY FINDINGS

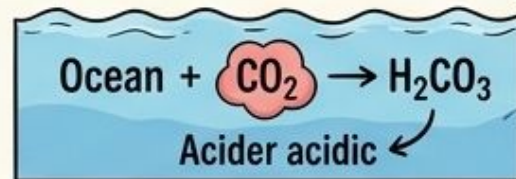
THE PLANETARY BOUNDARIES FRAMEWORK

(Concept: Johan Rockström)



DEEP DIVE: OCEAN ACIDIFICATION (THE 7th BREACH)

"Evil Twin of Climate Change"



Making water acidic

IMPACT ON CALCIFIERS



Stressed corals



Mollusks



Shell plankton

METRIC: Aragonite Saturation Levels
< 80% pre-industrial threshold

SOE 2026: INDIA-SPECIFIC FINDINGS

EXTREME WEATHER



EXTREME EVENTS ON
99% OF DAYS (2025 Data)

MONITORING GAPS
Only 15% Population
covered by monitors
within 10km radius



HUMAN-WILDLIFE CONFLICT
Tiger Landscapes | Habitat
Saturation



VIKSIT BHARAT @ 2047
Need shift from 'Reactive
Compensation' to
'Proactive Resilience'

SHORT NOTES FOR REVISION (Quick Facts)

NOVEL ENTITIES	GREEN WATER	PLANETARY HEALTH CHECK
Plastics, PFAS, Nuclear Waste, GMOs	Rain, Soil Moisture, Evaporation (distinct from distinct from Blue Water)	New Annual Scientific Review tracking these boundaries

UPSC CSE PRELIMS 2026: ENVIRONMENT TRENDS

ENVIRONMENT & ECOLOGY: PRELIMS 2026 FOCUS

1. INVASIVE SPECIES: LANTANA CAMARA

THE SPREAD & IMPACT

★ Introduced (19th Century) as ornamental; Now occupies ~50% of India's forest & scrublands.

🌍 **GLOBAL STATUS:** One of World's Top 10 Worst Invasives.

🌿 **ECOLOGICAL DEGRADATION:** Allelopathic chemicals suppress native grasses and shrubs.

🌱 **ECOLOGICAL DEGRADATION:** Allelopathic chemicals suppress native grasses and shrubs.

THE TIGER-LANTANA NEXUS: Suppresses native prey (Cheetal, Sambar) who migrate for fodder; Leads to Prey Depletion.



🐅 **INTENSIFYING HUMAN-TIGER CONFLICT:** Tigers use dense thickets as refuge outside protected areas; attack livestock near villages.

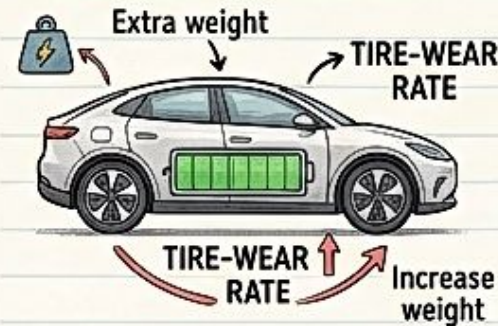
🌿 **MANAGEMENT:** Cut-Root Stock method; active restoration of native grasses.



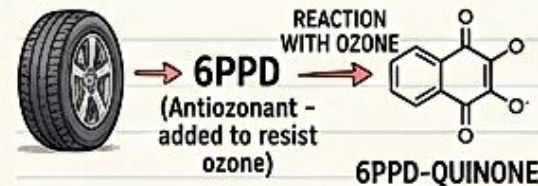
2. POLLUTION TRENDS: TIRES & PLASTICS



A. THE EV-TIRE PARADOX



THE TOXIC CASCADE



ACUTE TOXICITY to aquatic life

B. PLASTIC WASTE MANAGEMENT (AMENDMENT) RULES, 2026

➔ **KEY SHIFTS:** Towards Traceability & Circular Economy.

➔ **MANDATORY TRACEABILITY:** QR codes/Barcodes on packaging for producer-to-recycler tracking.

➔ **RECYCLED CONTENT TARGET:** Rigid plastic packaging must have minimum 40% recycled content by FY 2026-27.

➔ **EPR EXPANSION:** Targets for Extended Producer Responsibility now include 'Sellers' (raw material suppliers).

➔ **MICROPLASTICS STANDARDS:** Specific rules for defining and microplastics in industrial effluents.

➔ **ENFORCEMENT:** Shifted to Urban Local Bodies (ULBs) with digital auditing.

UPSC PRELIMS 2026: PROTECTED AREAS UPDATES



→ MADHAV TIGER RESERVE (MP)



India's 59th TR (2025)

- Major addition to India's tiger habitat
- Crucial corridor linking **PANNA TR (MP)** & **RANTHAMBORE TR (RJ)**
- Reintroduction success



KALI TIGER RESERVE (KARNATAKA)

Eco-Sensitive Zone (ESZ) Proposal (Early 2026)

- Proposed **663 sq km ESZ buffer**
(Regulated vs. prohibited activity)
- **Western Ghats** biodiversity hotspot
- Famous for **BLACK PANTHERS**



NEW RAMSAR SITES (India reaches 98, early 2026)

Spotlight on Key New Designations:

PATNA BIRD SANCTUARY (UP)

- Location: Etah district
- Smallest in UP
- Vital wintering ground for migratory birds (Central Asian Flyway)



CHHARI-DHAND WETLAND (GUJARAT)

- Location: Banni Grasslands, Kutch
- Ephemeral wetland (monsoon dependent)
- Home to **Common Crane**, **Dalmatian Pelican**, **Desert Fox**



KEY PRELIMS TAKEAWAYS

- Tamil Nadu has most Ramsar sites (20)
- Corridor connectivity focus
- ESZ function: Shock absorber

GLOBAL CLIMATE GOVERNANCE (FOR UPSC PRELIMS 2026)



UNFCCC COP29

(BAKU, AZERBAIJAN, 2024)

THE FINANCE COP



NEW COLLECTIVE QUANTIFIED GOAL (NCQG)

AIMING FOR
\$1.3 TRILLION
ANNUALLY BY 2035

- "Core" vs. nations to mobilize \$300 BILLION/year by 2035 as core public finance, with trolad/alternative sources

- "Core" vs. "Layer"
Developed nations to mobilize \$300 BILLION/year by 2035 as core public finance, with total including private/alternative sources)

\$1.3 Trillion/year
by 2035

Sources
Public
Private
Alternative

OPERATIONALIZED ARTICLE 6 (CARBON MARKETS)



Article 6.2
(Bilateral Trading)



Article 6.4
(UN-Backed Global Market)

BAKU HARMONIYA CLIMATE INITIATIVE



- Bridging gaps between climate finance & agriculture/food systems
- Empowering farmers



UNFCCC COP30

(BELÉM, BRAZIL, 2025)



IMPLEMENTATION AND FOREST COP

BELÉM POLITICAL PACKAGE

- focus on Tropical Forest Preservation & Global Implementation Tracker (monitoring fossil fuel transition pledge)



TROPICAL FORESTS FOREVER FACILITY (TFFF)

- Global fund paying countries for MAINTAINING standing forests (not just reforestation)



BELÉM HEALTH ACTION PLAN

- linking climate to public health resilience



BELÉM 4X PLEDGE

- quadruple global sustainable fuel use by 2035



UNCCD COP16

(RIYADH, SAUDI ARABIA, 2024)



UNCCD COP16: LAND RESTORATION & DROUGHT

- Discussions on RESTORING 1.5 BILLION HECTARES OF LAND BY 2030
- MANAGING GLOBAL DROUGHT



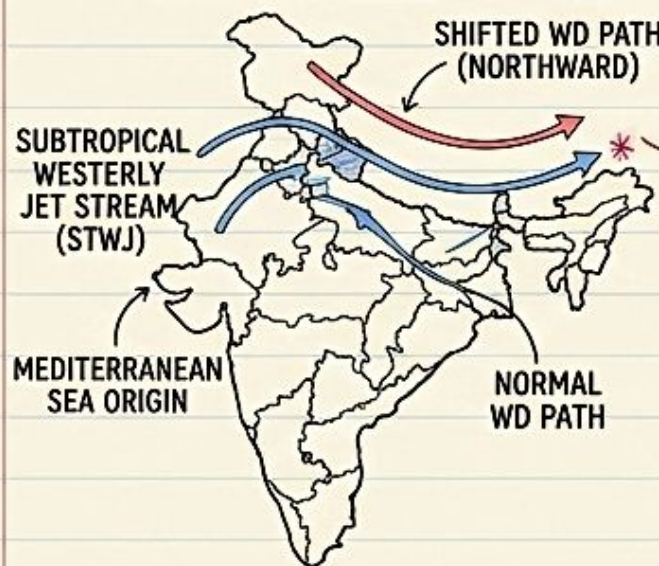
ABHISHA AI

by Abhishek Pathak

UPSC CSE PRELIMS 2026 | CLIMATOLOGY & INDIAN MONSOON TRENDS

1. SNOWFALL DEFICIT IN WESTERN HIMALAYAS (WINTER 2025-26)

WESTERN DISTURBANCES (WDs) PATH SHIFT



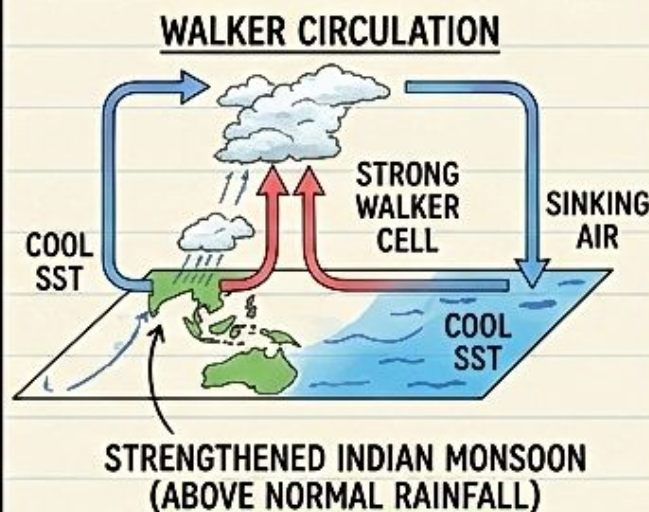
* **CAUSES:** Northward shift of STWJ, moisture-deficient WDs.

* **RESULT:** SNOWFALL DEFICIT IN W. HIMALAYAS

* **CONSEQUENCES:** Reduced glacier accumulation, worst reduced glacier accumulation, impact on perennial river flow, Rabi crops (wheat, mustard), and apple yield.

2. KEY INDIAN MONSOON DRIVERS

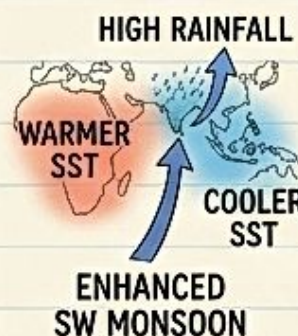
A. LA NIÑA (THE COOLING PHASE)



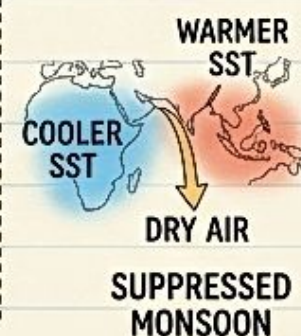
* Usually leads to more rain for India.

B. INDIAN OCEAN DIPOLE (IOD)

POSITIVE IOD (+IOD)



NEGATIVE IOD (-IOD)



HIGH-YIELD 'PRELIMS-STYLE' FACTS:

- WDs move WEST to EAST (Extra-tropical Cyclones).
- Worst monsoon years: El Niño + Negative IOD
- Global warming pushes snowline higher, reducing natural water banks.

3. COMPARATIVE SUMMARY FOR QUICK REVISION

PHENOMENON	NORMAL ROLE	2025-26 STATUS/TREND	POTENTIAL UPSC STATEMENT
Western Disturbances	Winter rain/snow for Rabi crops	<u>WEAK/NORTHWARD SHIFT</u>	Northward shift of STWJ leads to Himalayan snow deficit.
La Niña	Strengthens SW Monsoon	<u>EXPECTED TRANSITION</u>	La Niña usually causes a stronger Walker Circulation.
Positive IOD	Boosts Indian Rainfall	<u>VARIABLE</u>	Positive IOD is characterized by warmer SST in the Arabian Sea.
STWJ	Guides WDs into India	<u>LATITUDINAL SHIFT</u>	STWJ resides south of the Himalayas in winter.

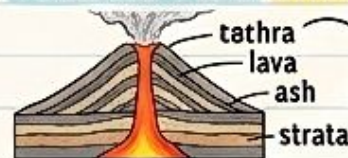
UPSC PRELIMS 2026 GEOMORPHOLOGY & VOLCANISM: ETNA & KILAUEA

12

MOUNT ETNA (SICILY, ITALY)

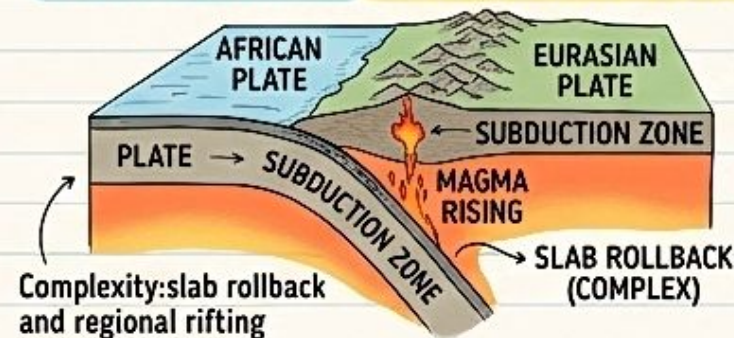
- KEY FACTS:** Europe's most active volcano
UNESCO World Heritage site

- VOLCANO TYPE:** STRATOVOLCANO (COMPOSITE VOLCANO)



Built of layers of hardened lava, tephra, pumice, ash.

- TECTONIC SETTING:** CONVERGENT PLATE BOUNDARY



- RECENT ACTIVITY (2025-26)**

- **Paroxysmal Events** (explosive bursts)

- **SEC Collapse:** Partial collapse of South East Crater in June 2025, leading to pyroclastic flows.



- **Valle del Bove:** Large horseshoe caldera, directs lava flows.

- **Strombolian Activity:** Mild explosions

RECENT ERUPTIONS
(JUNE 2025 - EARLY 2026)



STRATOVOLCANO
(Steep slopes)

COMPARATIVE ANALYSIS: ETNA VS. KILAUEA

Feature	Mount Etna	Kilauea
Volcano Type	Stratovolcano (Composite)	Shield Volcano
Magma Nature	Intermediate to Acidic (higher silica)	Basic/Mafic (low silica)
Eruption Style	Explosive & Effusive (Mixed)	Primarily Effusive (Fluid)
Plate Margin	Convergent (Subduction Zone)	Intra-plate (Hotspot)
Recent Feature	Pyroclastic flows, SEC collapse	Lava fountaining, Vog (SO ₂)



STRATOVOLCANO
(Steep slopes)

13

KILAUEA (HAWAII, USA)

LATE 2025 ERUPTION

- KEY FACTS:** One of world's most active volcanoes, on Big Island of Hawaii

- VOLCANO TYPE:** SHIELD VOLCANO



Broad, gentle slopes built by fluid BASALTIC LAVA.

- LAVA TYPE:** MAFIC/BASALTIC LAVA

- High fluidity, low silica, low viscosity

- Result: **EFFUSIVE ERUPTIONS**

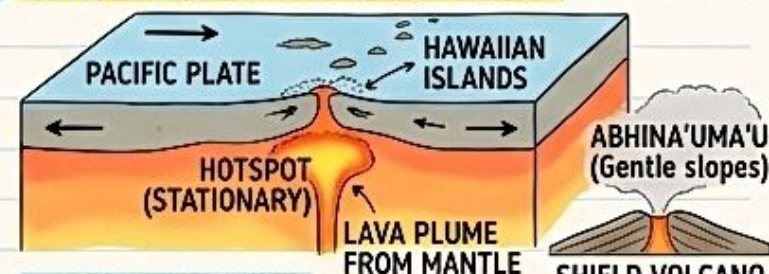


Pahoehoe (ropy)



'A'a (blocky)

- TECTONIC SETTING:** INTRA-PLATE HOTSPOT VOLCANISM



- RECENT ACTIVITY**

- Episode 38 & 39 centered on Halema'uma'u crater.
- Lava fountaining.

- ENVIRONMENTAL IMPACT**

VOG (VOLCANIC SMOG) - SO₂ + O₂ + H₂O with sunlight.

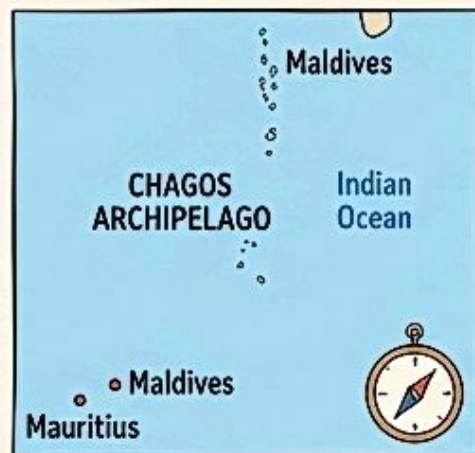
UPSC REVISION POINTS (STATIC & CURRENT)

- ☉ **The Ring of Fire** ☉ **Lava Viscosity** ☉ **Associated Landforms** ☉ **Envt. Impact**
- Focus, but note Etna in Med. belt
- Determines shield vs. stratovolcano
- Cinder cones
- Batholiths (intrusive)
- Calderas (extrusive)
- Ash and soil fertility (Andisols), cooling effect on climate

💡 **KEY POINT:** Etna = Convergent; Kilauea = Hotspot

IMPORTANT PLACES IN NEWS (STRATEGIC/ENVIRONMENTAL) FOR UPSC PRELIMS 2026

1 CHAGOS ARCHIPELAGO (Indian Ocean)



- **UK-MAURITIUS DEAL (MAY 2025/2026):** Historic treaty transfers sovereignty to Mauritius.
- **GEOPOLITICS:** UK cedes sovereignty; retains control over DIEGO GARCIA (major US-UK military base) for 99 YEARS.
- **STRATEGIC VALUE:** Key for Indian Ocean monitoring, Indo-Pacific operations.
- **ENVIRONMENTAL:** Establishment of Marine Protected Area (MPA) and resettlement for CHAGOSSIANs (displaced people).

2 JEJU ISLAND (South Korea)

- **UNESCO TRIPLE CROWN SITE:** Only place with all three:
 - ↳ BIOSPHERE RESERVE (2002)
 - ↳ WORLD NATURAL HERITAGE (2007) (Jeju Volcanic Island and Lava Tubes)
 - ↳ GLOBAL GEOPARK (2010)



GEOLOGICAL FEATURES:
HALLASAN (South Korea's highest shield volcano), Geomunoreum Geomunoreum Lava Tube System

- ↳ **UNIQUE CULTURE:** Haenyeo (female divers), Dol hareubang (stone statues)



- ↳ **UNIQUE CULTURE:** Haenyeo (female divers), Dol hareubang (stone statues)
- ↳ **TRAVEL ADVISORY** context (2026)

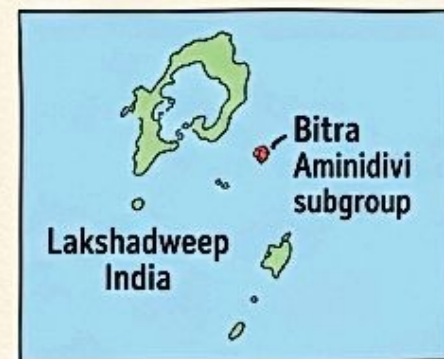
3 SENKAKU ISLANDS (East China Sea)



- **TERRITORIAL DISPUTE:** Japan (administers) vs China (claims as Diaoyu) vs Taiwan (claims as Diaoyutai).
- **STRAIN ESCALATION (Late 2025/2026):** Tensions over Japanese statements on Taiwan; Chinese "rights enforcement patrols".
- **STRATEGIC:** Key shipping lanes, potentially rich oil/gas reserves.
- **GEOGRAPHY:** Uninhabited islands, largest is Uotsuri Island.

↳ **NOT** part of the Nine-Dash Line

4 BITRA ISLAND (Lakshadweep, India)



- **SMALLEST INHABITED ISLAND** of Lakshadweep
- **STRATEGIC ACQUISITION (July 2025):** Govt initiates acquisition for defence.
- **DEFENSE OUTPOST:** Enhances Maritime Domain Awareness (MDA) with INS DWEEPRAKSHAK (Kavaratti) & Lamashen (Kavaratti) & INS JATAYU (Minicoy)
- **GEOGRAPHY:** Large lagoon (~45 sq km); Malik Mulla shrine.

↳ Note: sequence from North to South (Aminidivi, Laccadive, Minicoy)

REVISION TABLE: QUICK COMPARE

PLACE	REGION	KEY SIGNIFICANCE
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AGRICULTURE & NATURAL RESOURCES | PRELIMS 2026 FAST FACTS

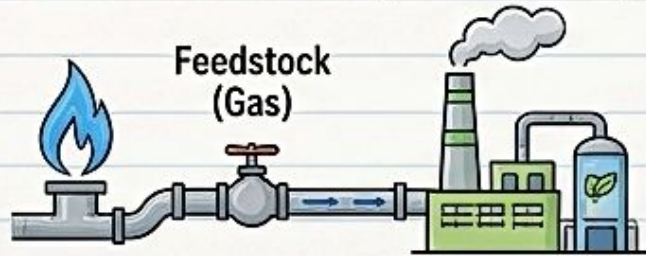


COLUMN 1



1. THE FERTILIZER CRISIS & INDIA

- 2026 West Asia conflict → 65% Urea price surge.
- India's 70% Import Dependency (Overall).
- UREA DYNAMICS: Statutory Controlled Price (MRP), Canalized Imports.
- GAS LINK: Natural Gas is primary Urea Feedstock.
- IMPACT: High subsidy burden; skewed NPK ratio (\$4:2:1\$ ideal).



- 'ONE NATION ONE FERTILIZER' & NANO UREA: Key solutions.

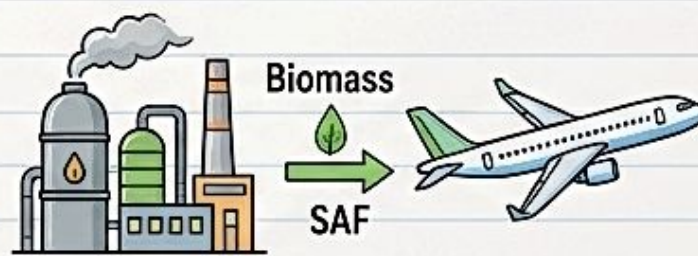


COLUMN 2



2. SUSTAINABLE AVIATION FUEL (SAF)

- A 'Drop-in' Fuel: Blend with ATF without engine change.
- 2026-27 MANDATE: 1% blending target by 2027.
- FEEDSTOCKS: Agricultural Residues, Corn Grain, Wood Mill Waste.
- FEEDSTOCK CATEGORIES: 1G (Edible - like corn/sugar) vs. 2G (Non-edible - like parali/crop waste). SAF focus is 2G.



- INTERNATIONAL FRAMEWORK: CORSIA (ICAO) mandatory from 2027.



COLUMN 3



3. LITHIUM EXPLORATION & KABIL

- KABIL: Joint Venture of NALCO, HCL, MECL (under Ministry of Mines).
- MISSION: Identify/acquire strategic minerals abroad.
- ARGENTINA (APRIL 2026): Environmental clearance for 5 brine-based blocks (Catamarca province).
- 'LITHIUM TRIANGLE': Argentina, Bolivia, Chile.



- EXTRACTION: Salar (salt flats) brine evaporation.
- DOMESTIC FIND: J&K Salal-Haimana (inferred G3 category).

